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A machine learning platform for genotype-specific cardiotoxicity risk prediction using patient-derived iPSC-CMs

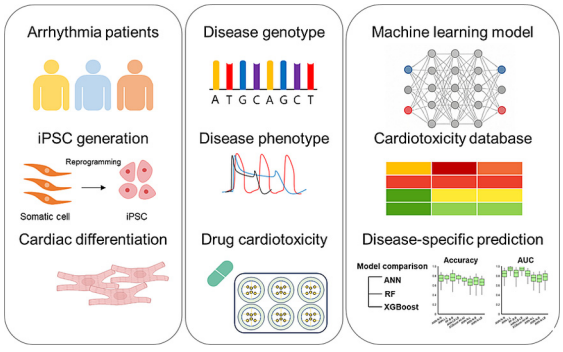
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HIGHLIGHTS

- Patient-specific responses to proarrhythmic drugs were observed in functionally validated iPSC-CMs.
- Arrhythmia-model iPSC-CMs exhibited increased sensitivity to drug-induced cardiotoxicity.
- Machine learning with patient-derived biomaterials enables precision cardiotoxicity screening.

GRAPHICAL ABSTRACT



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ABSTRACT

Introduction: Drug-induced Torsades de Pointes (TdP) has led to withdrawal of several drugs from the market. Individuals with inherited cardiac channelopathies are at increased risk due to their underlying electrophysiological vulnerability.

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Keywords:
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Objectives: We aimed to develop a machine learning (ML) platform for disease-specific cardiotoxicity using patient-specific induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs) combined with high-throughput microelectrode array (MEA) recordings.
Methods: We generated genetically confirmed and phenotypically characterized iPSC-CMs from patients with long QT syndrome (LQTS) and Brugada syndrome (BrS). These cells were exposed to 28 compounds with varying TdP risk levels. Electrophysiological responses including field potential duration, corrected field potential duration, beat period and amplitude were measured using MEA. These data were used to train and compare machine learning models, including artificial neural networks (ANN), random forest, and XGBoost. Model performance was optimized by grid search and evaluated by fivefold cross-validation.
Results: The ANN model trained on LQTS iPSC-CMs achieved the highest accuracy (area under the curve [AUC] = 0.94). BrS cell lines showed hypersensitivity to calcium channel blockers, while LQTS lines exhibited heightened responses to potassium channel inhibitors. Previously ambiguous compounds were reclassified based on disease-specific electrophysiological profiles, demonstrating the platform's utility in genotype-specific cardiotoxicity risk assessment.
Conclusion: This study presents a scalable and individualized approach for cardiotoxicity screening using well-characterized patient-derived iPSC-CMs. The platform enhances drug safety prediction, supports regulatory evaluation, and advances precision medicine in arrhythmia risk assessment.
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Introduction

Torsades de Pointes (TdP) is a rare but potentially life-threatening ventricular arrhythmia that remains a critical concern in the field of cardiac safety pharmacology [1,2]. Despite the existence of stringent regulatory frameworks, numerous non-cardiac drugs have been withdrawn from the market due to unanticipated proarrhythmic effects [3,4]. However, conventional preclinical strategies, predominantly hERG channel blockade assays and QT interval measurements in animal models, are inadequate in capturing the physiological complexity and patient specificity necessary for accurate prediction of TdP risk in humans [5,6].

Human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) have emerged as a promising model for cardiac safety evaluation owing to their ability to recapitulate patient-specific electrophysiological features [6]. Numerous studies have demonstrated the utility of these cells in evaluating ion channel function and drug-induced arrhythmic effects [7–9]. In addition, microelectrode array (MEA) systems offer a high-throughput, non-invasive platform for quantifying these responses [6]. In parallel with these developments, machine learning (ML) has shown considerable promise in predicting cardiotoxicity by identifying non-obvious patterns in high-dimensional electrophysiological datasets [10]. Nevertheless, the integration of disease-specific genetic backgrounds into predictive models, and the subsequent interpretation of how it influences the electrophysiological response to drug treatment, have been addressed in only a limited number of approaches [10].

Despite these advances, most current models remain limited in their ability to incorporate disease-relevant genetic variation into predictive frameworks. This limitation is of particular consequence for individuals with mutations linked to inherited arrhythmia syndromes, such as long QT syndrome (LQTS) and Brugada syndrome (BrS), who exhibit heightened vulnerability to torsadogenic drugs [11]. Even so, existing experimental models and predictive tools are predominantly derived from healthy donor-derived cardiomyocytes and thus do not accurately reflect the electrophysiological heterogeneity introduced by disease-specific features [12,13]. Although recent studies have applied ML to analyze iPSC-CM data, most rely on narrow feature sets, offer limited interpretability, and rarely incorporate disease-specific responses [10,14].

To overcome these limitations, a ML–enabled platform was developed that integrates multichannel MEA recordings with iPSC-CMs from individuals with LQTS and BrS, as well as from unaffected controls. The workflow included characterization of both genetic mutations and arrhythmia-related electrophysiological abnormalities in iPSC-CMs. Only cell lines with confirmed genetic mutation and disease-relevant functional phenotypes were exposed to a panel of 28 clinically relevant compounds with varying TdP risk levels. This approach enabled the training of ML models capable of accurately stratifying arrhythmogenic risk (area under the curve [AUC] = 0.94) and identifying disease-dependent electrophysiological signatures. The findings of this study demonstrate distinct hypersensitivity patterns, such as enhanced responses to calcium channel blockers in BrS lines. The study establishes a scalable framework for individualized cardiotoxicity risk prediction. This platform offers broad utility in preclinical drug development and precision safety screening.

Importantly, this platform could serve as a practical tool for early drug screening and personalized safety assessment in genetically at-risk populations. The integration of disease-specific *in vitro* models with ML has the potential to improve the preclinical drug evaluation pipeline and support regulatory decisions aimed at minimizing cardiotoxic risk.

Methods

Donor cell preparation

Among the patients diagnosed with inherited arrhythmias enrolled in the Korean Inherited Arrhythmia Registry, four with LQTS and four with BrS, all of whom had strong evidence of inherited arrhythmias, were selected and followed up at outpatient clinics of the Seoul St. Mary's Hospitals. Each individual had a prolongation of the corrected QT interval (QTc $\geq$ 460 ms) and/or documented TdP, ventricular fibrillation, or cardiac arrest. All patients underwent a complete clinical evaluation, including electrocardiography (ECG), transthoracic echocardiogram, 24-h ECG Holter recording, and exercise monitoring.

Blood samples were obtained from these patients. All participants were Korean, and both males and females were included in the study. Sex-specific information for each cell line is provided in the accompanying table (Fig. 1A). Written informed consent was obtained from all the patients and their family members. This study was conducted in accordance with the ethical guidelines of the Declaration of Helsinki and approved by the Institutional Review Board/Ethics Committee of Seoul St. Mary's Hospital (No. KC16TNMI1023).

A

Disease	Cell line	Sex	Age	Mutated gene	Nucleotide change	Amino acid change	References / ClinVar
Normal	CMC-hiPS-011	Male	-	-	-	-	[15]
	PB01-EiPS21	Male	50	-	-	-	[16]
	TnR-hiPS-4	Female	19	-	-	-	[17]
LQTS	DPHCi01	Female	55	KCNH2	c.453delC	p.Thr152fs	[22-24]
	DPHCi08	Female	27	KCNQ1	c.569G>A	p.Arg190Gln	[25]
	DPHKi20	Male	20	KCNQ1	c.464T>G	p.Leu155Arg	NM_181798.1
	DPHCi02	Male	76	KCNH2	c.3107_3112 dupGCGACG	p.Gly1036_Asp 1037dup	NM_000238.3
BrS	DPHCi15	Male	38	SCN5A	c.1153G>A, c.1511G>C	p.Ala385Thr, p.Arg504Thr	[26]
	DPHKi09	Male	26	ANK2	c.4600T>G	p.Tyr1534Asp	NM_020977.3
	DPHKi08	Male	50	KCNE1	c.253G>A	p.Asp85Asn	[27,28]
	DPHKi11	Male	31	CACNB2b	c.1513A>G	p.Thr505>Ala	NM_201590

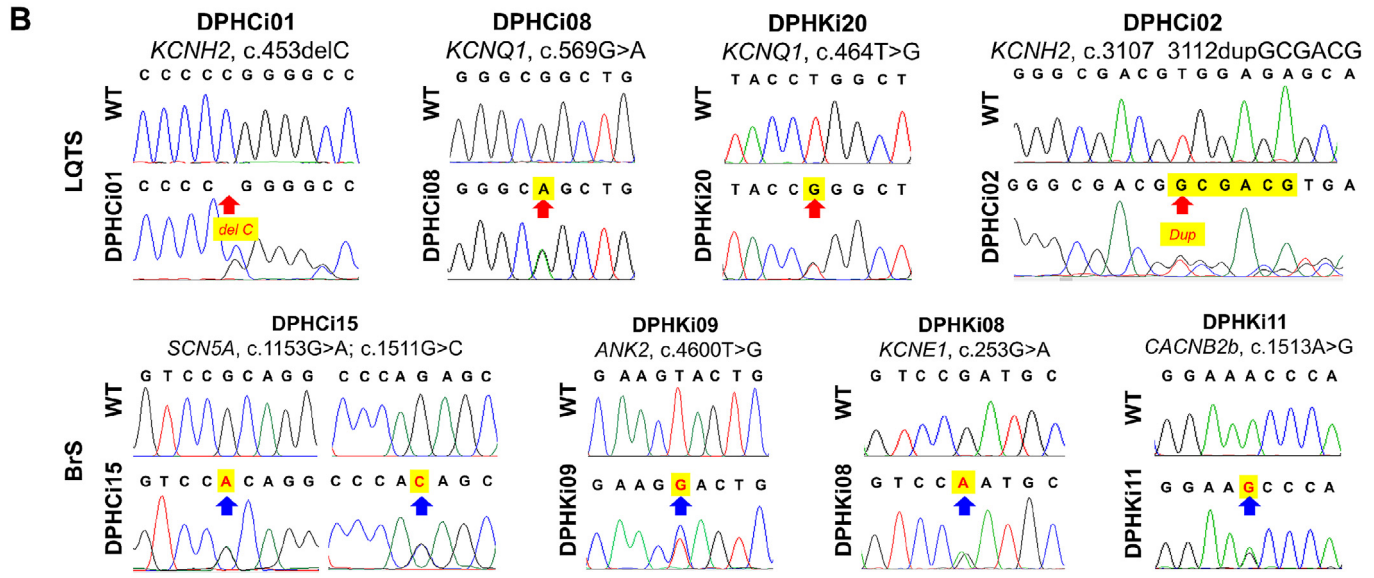


Fig. 1. Generation and characterization of human induced pluripotent stem cell from patients with inherited arrhythmias. (A) Donor and clinical information of normal controls and patients with long QT syndrome (LQTS) or Brugada syndrome (BrS). Peripheral blood mononuclear cells (PBMCs) from LQTS and BrS patients were isolated and subsequently reprogrammed into induced pluripotent stem cells (iPSCs). (B) Genetic mutations identified in patient-derived iPSCs. The upper panel depicts mutations in potassium channel genes (*KCNH2* and *KCNQ1*) from patients with LQTS, including deletions, base substitutions, and duplications (red arrows). The lower panel illustrates base substitutions in *SCN5A*, *KCNE1*, *CACNB2b*, and *ANK2*, associated with patients with BrS (blue arrows). (C) Representative images of iPSC colony morphology and karyotype analysis for patient-derived iPSC lines. All lines exhibited normal colony morphology and stable karyotypes, confirming the quality of the reprogrammed iPSCs. Scale bar = 500 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Generation and maintenance of iPSC lines

Human peripheral blood mononuclear cells donated by patients with a genetic arrhythmic disease were cultured in StemSpan™ SFEM II medium (STEMCELL Technologies, Vancouver, BC, Canada) and reprogrammed to iPSCs using a CytoTune™-iPS 2.0 Sendai Reprogramming Kit (Thermo Fisher Scientific, Waltham, MA, USA), according to the manufacturers' instructions. Full protocols

are described in the Supplementary Methods. The three normal iPSC lines (CMC-hiPSC-011, PB01-EiPS21, and TnR-hiPS-4F) were obtained from previous studies [15–17].

Mutation analysis

Genomic DNA was extracted using a Maxwell® RSC Blood DNA Kit (Promega, Madison, WI, USA), and Sanger sequencing was performed using designed primers (Table S1).

Karyotyping analysis

Karyotyping was performed using standard cytogenetic procedures and the GTG-band method with 500-band resolution at Samkwang Medical Laboratories (Seoul, Republic of Korea).

Cardiomyocyte differentiation

The hiPSCs were differentiated into cardiomyocytes as previously reported [17–19]. Spontaneous beating cells were purified and maintained. Full details are available in the Supplementary Methods.

Quantitative reverse transcription-polymerase chain reaction (qRT-PCR)

Total RNA from hiPSC-CMs was isolated using TRIzol™ Reagent according to the manufacturer's instructions. Complementary DNA (cDNA) was synthesized using a high-capacity cDNA reverse transcription kit. qRT-PCR was performed using FastStart Essential DNA Green Master Mix on a Light Cycler 96 system (Roche, Basel, Switzerland). All primer pairs were designed using Primer 3. The primer sequences for *GAPDH*, *TNNT2* and *MYH6* genes are listed in Table S2. Relative gene expression was calculated based on the average cycle threshold (Ct) value for technical triplicates, normalized to that of the *GAPDH* control, and compared with that in iCell CMs using the $2^{-\Delta\Delta Ct}$ method.

Immunofluorescence

Immunofluorescence analysis was performed for Troponin T2 (TNNT2) and sarcomeric-α-actinin to confirm cardiomyocyte identity. Antibody source and staining conditions are described in the Supplementary Methods (Table S3).

Flow cytometry

The hiPSC-CMs were resuspended and fixed with a BD Cytofix/Cytoperm™ Fixation/Permeabilization Solution Kit (BD biosciences, Franklin Lakes, NJ, USA) for 15 min, stained with Alexa Fluor® 647 Mouse Anti-Cardiac Troponin T (1:50) for 30 min at 4°C, and analyzed using a SH800S Cell Sorter flow cytometer with Cell Sorter v2.1.5 (Sony Biotechnology, San Jose, CA, USA).

Action potential (AP) recording and analysis

The APs of hiPSC-CMs were recorded using a whole-cell current-clamp configuration (Molecular Devices, San Jose, CA, USA). Classification criteria are outlined in the Supplementary Methods.

MEA data recording

To record their drug response, hiPSC-CMs were first seeded in 96-well MEA plates (Cytoview; Axion BioSystems, Atlanta, GA, USA). Field potentials were recorded by spontaneously contracting hiPSC-CMs using an MEA system (Maestro Pro MEA; Axion BioSystems). Concentration and exposure time details are available in the Supplementary Methods.

Data selection and augmentation

Electrophysiological data from hiPSC-CM lines exposed to 28 drugs at four concentrations were curated to evaluate TdP risk. MEA biomarkers—including field potential duration (FPD), corrected FPD using Fridericia's formula (FPDc), beat period, and amplitude—were averaged across doses to reduce variability and enhance robustness. Drug risk levels were annotated based on prior classifications [2]. Details of concentration selection, normalization criteria, and data augmentation procedures are described in the Supplementary Methods.

ML algorithms and grid search for optimal prediction

An artificial neural network (ANN) was developed using the Keras framework with four MEA biomarkers as input and binary TdP risk as output (Fig. 6B) [20]. Model architectures for ANN, random forest (RF), and XGBoost were optimized via grid search with five-fold cross-validation. The ANN outperformed other models in classification accuracy. Supplementary Methods provide full details on the model structure and parameter ranges.

ML testing mechanism and performance evaluation

We implemented various testing scenarios to assess the model's predictive performance across disease-specific cell lines [21]. In each group (normal, LQTS, BrS), 11 compounds were used for training and 15 for testing (Fig. 6A). Cross-condition testing evaluated generalizability by applying models trained on one genotype to another. Additional scenarios included prediction of novel drug responses and focused testing using subsets of normal and arrhythmic cell types. Classification accuracy and other performance metrics were derived from 10,000 test simulations. Full evaluation strategies and metric definitions are described in the Supplementary Methods.

Statistical analysis

Data are expressed as the mean ± standard deviation (SD). Unpaired two-tailed Student's *t*-tests were used to assess the differences between groups. All statistical analyses were conducted using GraphPad Prism (GraphPad Software), with statistical significance defined as $p < 0.05$.

Results

Generation of iPSCs from patients with inherited arrythmias

To generate inherited arrythmia iPSC lines, we perform genetic mutation screening and reprogramming of somatic cells (Fig. 1). The study population consisted of eight patients, two males and two females, aged 20–76 years, diagnosed with LQTS; and four males, aged 26–50 years, diagnosed with BrS (Fig. 1A). The peripheral blood mononuclear cells were isolated from blood sample and genetic mutations were examined by Sanger sequencing (Fig. 1A and 1B). Cells from all patients exhibited mutations in ion channels or related genes that could potentially influence cardiac APs (Fig. 1B).

Deletions, base substitutions, and duplications were identified in the potassium channel genes, *KCNH2* and *KCNQ1*, in cells derived from patients with LQTS (Fig. 1B). Specifically, a frameshift mutation resulting from a cytosine deletion at position 453 of

KCNH2 was identified in the DPHCi01 cell line (NM_000238.4) [22–24]. Additionally, a guanine-to-adenine base substitution at position 569 of *KCNQ1* was identified in the DPHCi08 cell line (NM_000218.3) [25] and a thymine-to-guanine base substitution at position 464 of *KCNQ1* was identified in the DPHKi20 cell line (NM_181798.1). The DPHCi02 cell line exhibited a duplication of bases 3107–3112, GCGACG, within *KCNH2* (NM_000238.3; Fig. 1B). Cells derived from patients with BrS exhibited base substitutions in *SCN5A*, *KCNE1*, and *CACNB2b*, which encode Na^+, K^+, and Ca^{2+} channels, respectively, or in *ANK2*, involved in the stabilization of ion channel proteins (Fig. 1B). Specifically, two base substitutions were identified in the DPHCi15 cell line: 1153 guanine to adenine and 1511 guanine to cytosine in *SCN5A* (NM_001160161.1) [26]. A single base substitution was identified in the DHPKi09 cell line: 4600 thymine to guanine in *ANK2* (NM_020977.3). The DPHKi08 cell line exhibited a base substitution of 253 guanine to adenine in *KCNE1* (NM_000219.6) [27,28], while the DPHKi11 cell line displayed a base substitution of 1513 adenine to guanine in *CACNB2b* (NM_201590; Fig. 1B).

Human iPSC lines were generated by inducing the reprogramming of cells derived from patients with the identified gene mutations. The cells proliferated as typical iPSC colonies (Fig. 1C), and the pluripotency was confirmed by gene and protein levels (data not shown). The genetic stability of the reprogramming process verified that all eight cell lines exhibited normal karyotypes (Fig. 1C). This process yielded eight iPSC lines characteristic of inherited arrhythmia, each with a distinct set of gene mutations.

Cardiomyocyte differentiation and disease-specific baseline electrophysiological profiles

To generate cardiomyocytes from normal control and inherited arrhythmia iPSC lines (Fig. 2), we used a previously established small molecule-based 2D differentiation method (Fig. 2A) [17–19]. Following a 3–4-day culture to achieve 90 % confluence, iPSCs were treated with CHIR99021 and Wnt-C59 to induce differentiation. Spontaneously beating cardiomyocytes were observed after 8–10 days of differentiation (Fig. S1 and Video S1–S11). The cardiomyocytes were subsequently purified, cultured for 14 days, and subjected to characterization and drug response assays. Marker expression in beating cells was evaluated using qRT-PCR and immunostaining techniques. Comparison with global control iCell cardiomyocytes revealed that *TNNT2* exhibited 1.73–2.49-fold increases in expression in normal cell lines, 0.49–2.25-fold in LQTS cell lines, and 1.01–2.23-fold in normal cell lines (Fig. 2B). Myosin Heavy Chain 6 (*MYH6)* exhibited a 0.42–2.30-fold upregulation in normal cell lines, 0.92–3.99-fold in LQTS cell lines, and 1.05–5.04-fold in BrS cell lines (Fig. 2B). Staining for the intracellular cardiomyocyte-specific markers TNNT2 and sarcomeric α-actinin showed that this protein was strongly expressed in all cell lines (Fig. 2C). The proportion of TNNT2-positive cells was quantified using flow cytometry to evaluate the efficiency of cardiomyocyte production in each cell line.

All cell lines exhibited cardiomyocyte production efficiencies exceeding 90 % (normal cell line: 91.1–96.6 %, LQTS cell line: 93.2–97.6 %, BrS cell line: 92.2–98.9 %; Fig. 2D). These findings indicated that the established cardiomyocyte differentiation methodologies generated high-purity cardiomyocytes from iPSCs derived from both healthy individuals and patients with inherited arrhythmias.

Characterization of inherited arrhythmia cardiomyocytes through electrophysiological function analysis

To investigate the electrophysiology of the iPSC-CMs from healthy control and arrhythmia patients, we recorded the action potential parameter by patch clamp analysis (Fig. 3). The proportions of ventricular-, atrial-, and nodal-like cells were determined for each cell line (Fig. S2). In normal cell lines, ventricular-like cells constituted 70.5 ± 1.25 %, atrial-like cells 20.3 ± 3.11 %, and nodal-like cells 9.3 ± 3.66 %. LQTS cell lines exhibited 59.8 ± 3.00 % ventricular-like, 28.8 ± 7.17 % atrial-like, and 11.3 ± 5.68 % nodal-like cells. In contrast, BrS cell lines showed significantly reduced ventricular-like proportions (29.8 ± 35.95 %) and increased atrial-like proportions (65.4 ± 34.76 %; Fig. 3A). Notably, no ventricular-like APs were observed in the BrS cell lines DPHCi15 and DPHKi09, where 100 % and 87.5 % of the cells exhibited atrial-like APs, respectively. To determine whether the DPHCi15 and DPHKi09 cell line-derived cardiomyocytes consisted exclusively of atrial-type cells, we performed qPCR analysis of the ventricular and atrial markers, Myosin Light Chain 2 (*MYL2*) and Myosin Light Chain 7 (*MYL7),* respectively. *MYL2* expression in DPHCi15 and DPHKi09 cells was comparable to that in iCell cardiomyocytes, whereas *MYL7* expression was lower in both cell lines compared to that in iCell cardiomyocytes. These results indicate that the cells were not specifically directed toward atrial-type differentiation (Fig. S3). Rather, the atrial-like electrophysiological properties observed are presumed to reflect intrinsic BrS-associated characteristics.

Normal cell lines exhibited typical ventricular-like APs with normal APD_{80} values (Fig. 3B). In LQTS cell lines (DPHCi01, DPHCi08, and DPHKi20), prolonged APD_{80} values (587 ± 181, 308 ± 17.7, and 556 ± 70.9 ms, respectively) were observed, consistent with the characteristics of LQTS (Fig. 3B). However, the DPHCi02 line exhibited AP characteristics comparable with those of normal cell lines, despite harboring genetic mutations associated with LQTS.

For BrS cell lines, phase 2 loss was identified in the AP recordings of DPHCi15 and DPHKi09, resembling the epicardial dome-loss phenomenon typical of BrS [29]. In contrast, DPHKi08 and DPHKi11 exhibited AP characteristics similar to those of normal cell lines, without the hallmarks of BrS (Fig. 3B).

Quantitative analysis revealed significantly prolonged APD_{80} in the three LQTS cell lines (DPHCi01, DPHCi08, and DPHKi20) compared with that of normal cell lines (Fig. 3C). Conversely, the BrS cell lines DPHCi15 and DPHKi09 displayed shortened AP amplitude (80.3 ± 2.7, 82.9 ± 2.4 mV) and APD80 (119 ± 10.2, 94.1 ± 7.1 ms), consistent with BrS phenotypes (Fig. 3C).

Based on these findings, three LQTS (DPHCi01, DPHCi08, and DPHKi20) and two BrS (DPHCi15 and DPHKi09) cell lines were selected for subsequent experiments to model the disease characteristics of iPSC-CMs. Additionally, the absence of inherited arrhythmia phenotypes in certain cell lines (e.g., DPHCi02, DPHKi08, and DPHKi11), despite the presence of validated genetic mutations, highlights the necessity of associating disease phenotypes at the cellular level when developing iPSC-based models.

Analysis of drug-induced changes in cardiomyocytes from healthy individuals and patients with inherited arrhythmias

To identify the cardiotoxic effects of drugs in iPSC-CMs, the MEA technique was used to measure real-time electrophysiological functions in response to drug treatment in nine cardiomyocyte cell lines: four normal, including iCell cardiomyocytes, and five (LQTS: three, BrS: two) with inherited arrhythmia features (Fig. 4). The MEA technique measures the field potential generated by a contracting cardiomyocyte on an electrode through four parameters (Fig. 4A), including FPD (the interval between depolarization to repolarization of the FP), FPDc, beat period (BP, interval between one beat and the subsequent beat), and amplitude (potential difference between the highest and lowest potentials in the FP cycle). In

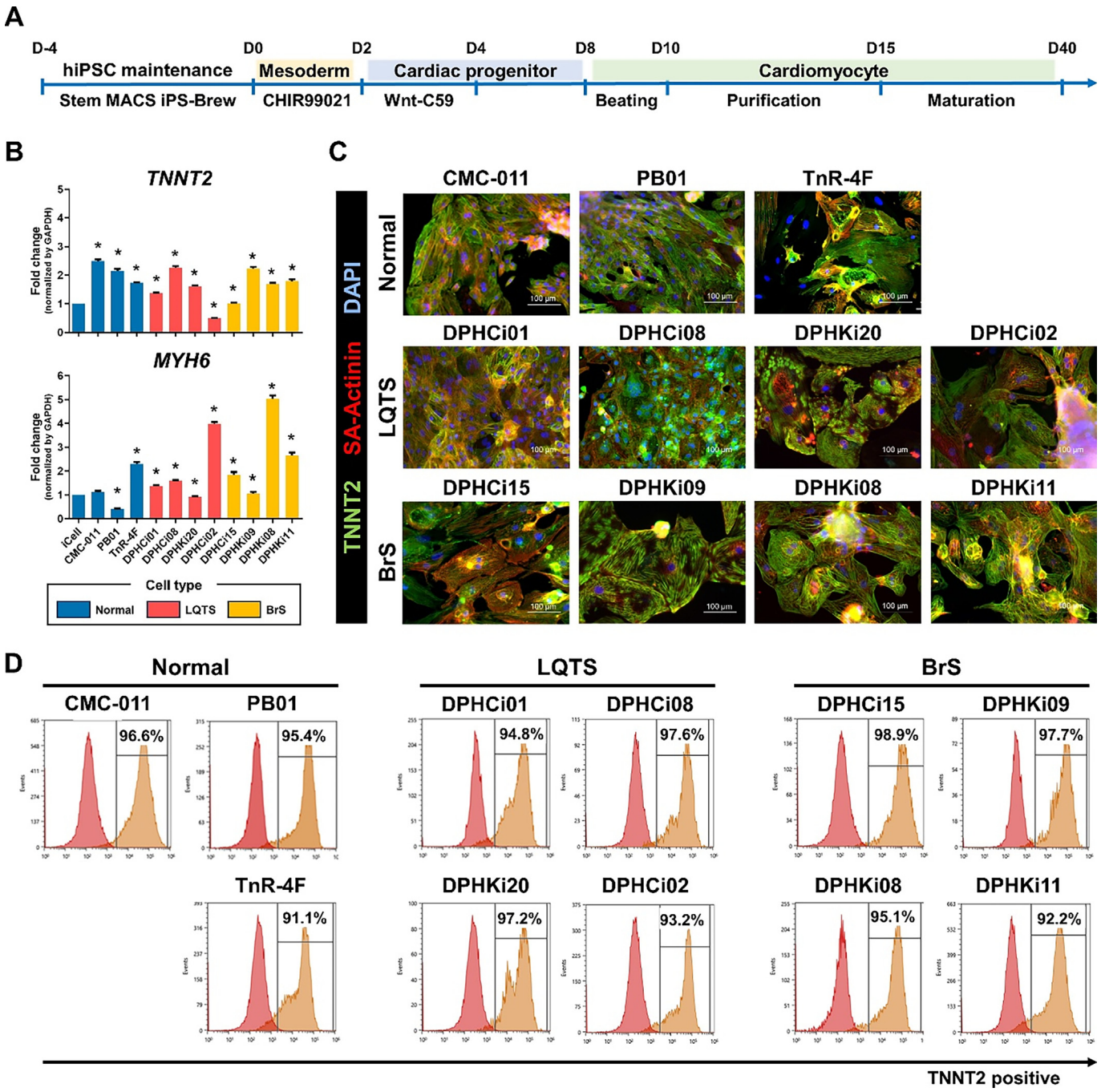


Fig. 2. Differentiation and characterization of cardiomyocytes (CMs) derived from normal and inherited arrhythmia iPSC lines. (A) Schematic representation of the 2D differentiation protocol used to generate CMs from iPSC lines. (B) Expression levels of CM-specific genes, *TNNT2* and *MYH6*, in normal, LQTS, and BrS iPSC-CMs, relative to global control iCell CMs. Data represents mean ± SD. $^{*}p < 0.05$ compared with iCell cardiomyocytes; $n = 3$. Student's *t*-test was used for statistical analyses. (C) Immunostaining of TNNT2 (green) and sarcomeric α-actinin (red) in normal, LQTS, and BrS iPSC-CMs, demonstrating strong intracellular expression of CM-specific markers. Nuclei were counterstained with DAPI (blue). Scale bar = 100 μm. (D) Flow cytometry analysis of TNNT2-positive cells in normal, LQTS, and BrS CM lines. All cell lines exhibited high CM production efficiencies exceeding 90 %, with normal cell lines achieving 91.1–96.6 %, LQTS lines 93.2–97.6 %, and BrS lines 92.2–98.9 %. TNNT2; Troponin T2, *MYH6*; Myosin Heavy Chain 6. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the MEA analysis, cardiomyocyte FP is observed as either a continuous or single-beat morphology (Fig. 4A).

Twenty-six drugs were selected for this study based on the representative compound list proposed by the Comprehensive *in Vitro* Proarrhythmia Assay (CiPA) initiative [30]. This list classifies compounds according to their known proarrhythmic risks (Table 1). Most of these drugs are listed in CredibleMeds® as having a risk of TdP [31]. Indeed, several have been withdrawn from the market or received regulatory warnings due to their cardiotoxic and/or arrhythmogenic effects [32,33]. Twelve of these compounds act as blockers of cardiac K^+, Na^+, or Ca^{2+} ion channels, influencing cardiac action potentials. Several others target membrane receptors, including five that bind to dopamine receptors, three to histamine receptors, and three to serotonin receptors. Other drugs include tyrosine kinase inhibitors, antibiotics, and estrogen receptor ligands.

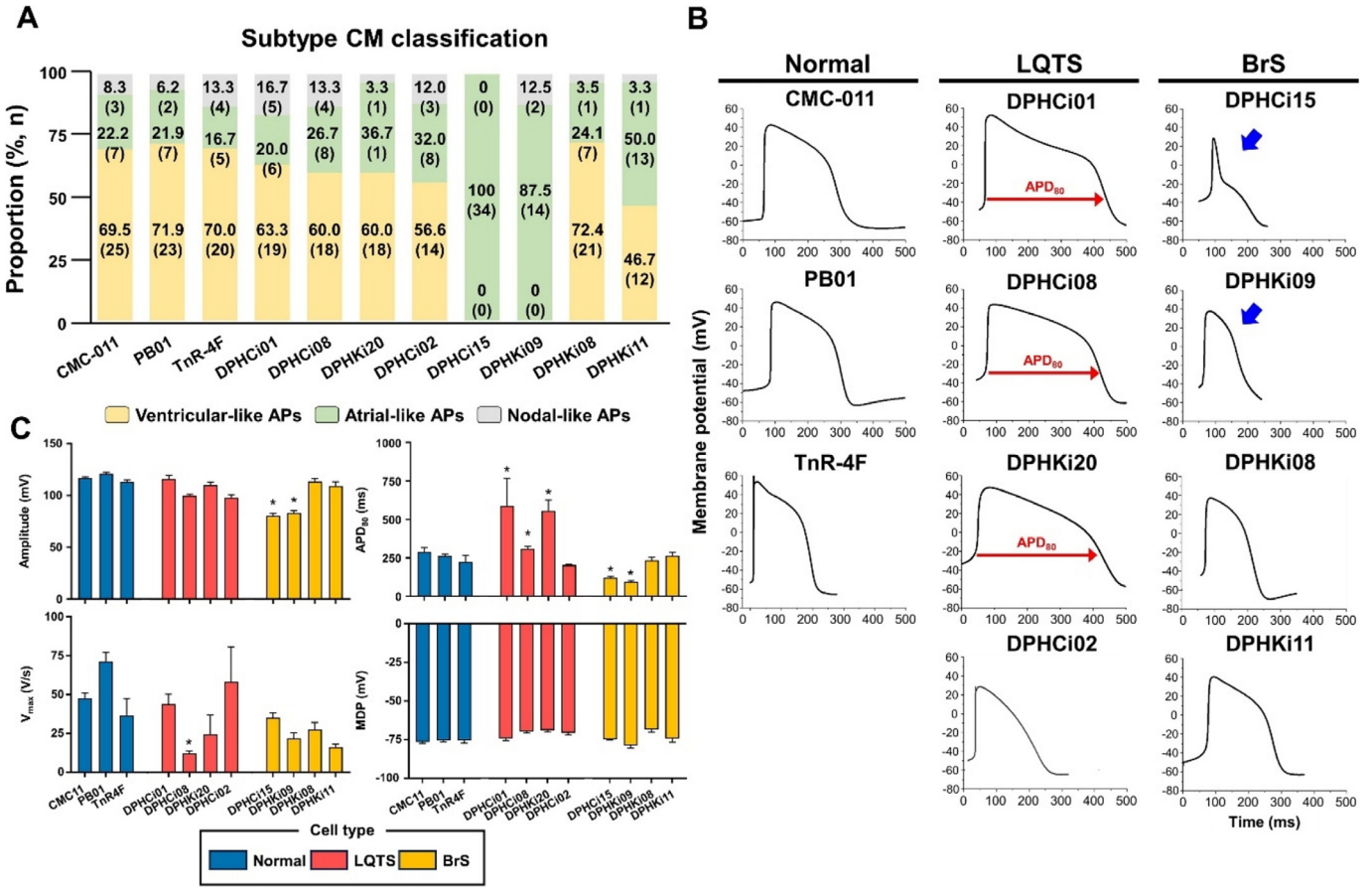


Fig. 3. Electrophysiological properties of cardiomyocytes (CMs) derived from normal and inherited arrhythmia iPSC lines. (A) Subtype classification of ventricular-, atrial-, and nodal-like action potential (AP) in CMs derived from normal, LQTS, and BrS iPSC lines (n ≥ 16). (B) Representative AP traces. Prolonged action potential duration (APD) 80 in LQTS iPSC-CMs is indicated by red arrows. Dome-loss phenomena in BrS iPSC-CMs are marked by blue arrows. (C) Quantitative analysis of AP parameters in ventricular-like APs, including amplitude, APD80, V_{max} and MDP. The red line represents the mean value of normal cell lines. For the DPHKi15 and DPHKi09 cell lines, all recorded APs were analyzed, as no ventricular-like APs were observed in these lines. The data are represented as mean ± standard error of the mean (SEM). $^{*}p < 0.05$ compared with normal cell lines; $n \geq 12$ per cell lines. For statistical comparison, values from each disease cell line were compared against the pooled mean of three normal cell lines (CMC-011, PB01, TnR-4F). Student's *t*-test was used for statistical analyses. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Analysis of changes in normal and inherited arrhythmic cardiomyocytes following treatment with ion channel blockers for cardiovascular disease

The impact of 12 ion channel blockers with classified TdP risk was evaluated using MEA in normal, LQTS, and BrS iPSC-CMs. Changes in FPDc, a key parameter linked to the QT interval and cardiotoxicity risk, were measured before and after drug treatment [34]. In cases where drug treatment led to complete cessation of beating (quiescence), FPDc was no longer measurable. For analysis consistency, these conditions were represented as a 100 % change in FPDc to reflect the total loss of electrical activity, and are annotated with "Q" in the supplementary dataset.

K^+ channel blockers, azimilide and sotalol, induced concentration-dependent FPDc prolongation across all cell lines, with similar sensitivity observed in normal and inherited arrhythmia lines (40–80 % change at 1 μM; Fig. 4B).

Na^+ channel blocker disopyramide also prolonged FPDc across cell lines (Fig. 4C), while the low risk mexiletine caused minimal changes (<20 %) except in the DPHCi08 line (red arrow in Fig. 4C).

Ca^{2+} channel blockers showed enhanced sensitivity in BrS lines (Fig. 4D), particularly to nifedipine and nitrendipine, where FPDc prolongation and electrical instability were noted at low concentrations (blue arrow in Fig. 4D). Diltiazem exhibited similar effects in BrS lines, indicating their heightened vulnerability to Ca^{2+} blockers compared to normal and LQTS lines.

Among multichannel blockers, ibutilide caused pronounced FPDc changes in BrS and normal lines (>80 % at 0.01 μM) but exhibited lower sensitivity in LQTS lines (<32 %; Fig. 4E). Ranolazine (low TdP risk) showed higher sensitivity in BrS lines, with DPHCi15 exhibiting > 40 % FPDc change at 10 μM (purple arrow in Fig. 4E). Quinidine (high TdP risk) resulted in greater FPDc changes in LQTS and BrS lines (>40 % at 3 μM) than that in normal lines (<30 %; black arrow in Fig. 4E).

These findings highlight the distinct electrophysiological responses of BrS and LQTS iPSC-CMs to ion channel blockers, emphasizing the utility of MEA-based assays for evaluating cardiotoxicity and disease-specific drug sensitivities in arrhythmic models.

Analysis of normal and inherited arrhythmic cardiomyocyte responses to treatment with non-cardiovascular drugs

The effects of 14 non-cardiovascular drugs with reported cardiotoxicity were evaluated in iPSC-CMs using MEA analysis (Fig. 5, Data set S1). Compared to ion channel blockers, non-cardiovascular drugs generally induced minimal FPDc changes.

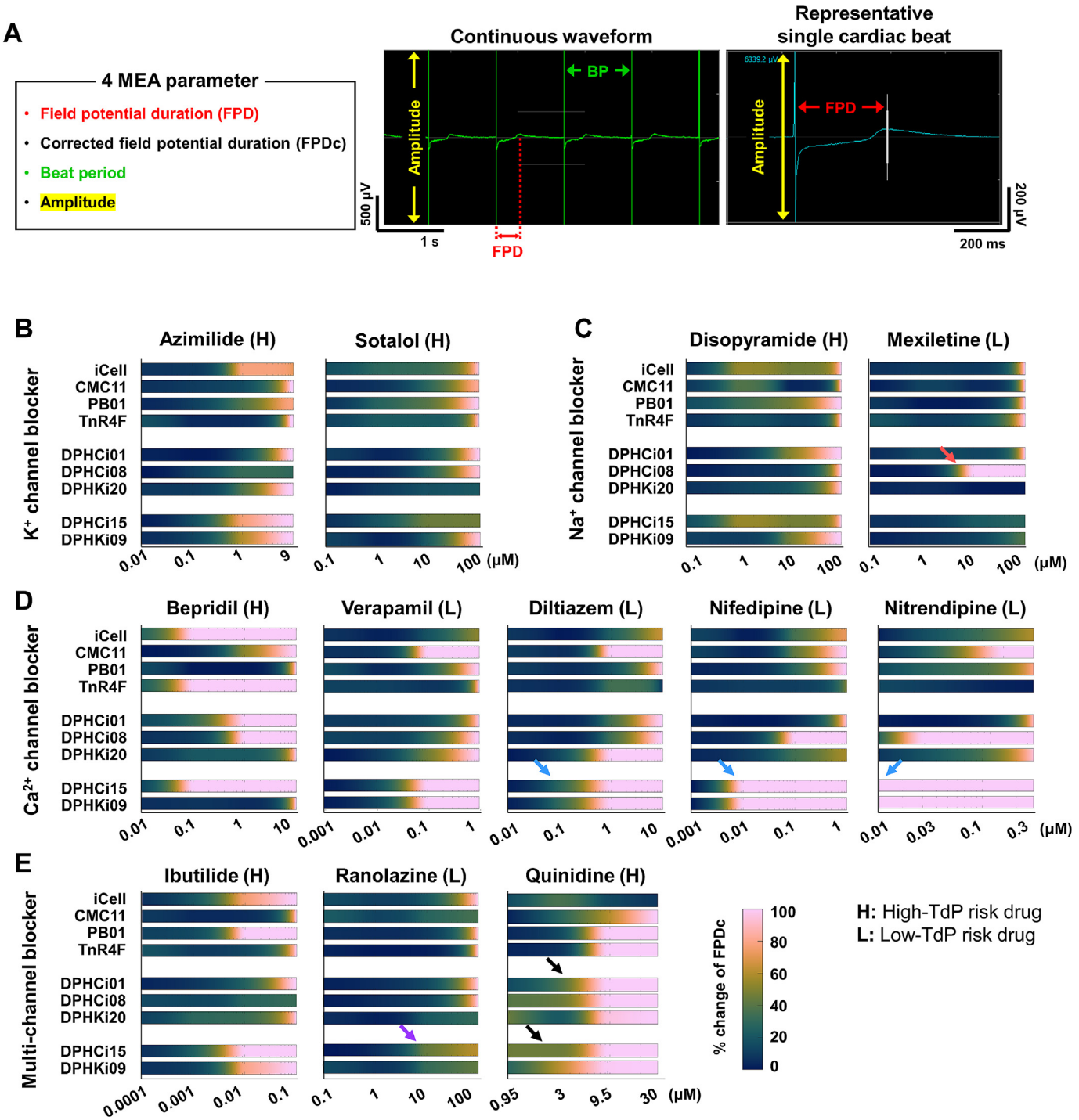


Fig. 4. Drug-induced electrophysiological changes in iPSC-derived cardiomyocytes (iPSC-CMs) analyzed using a microelectrode array (MEA). (A) MEA parameters used to evaluate iPSC-CMs. Field potential duration (FPD, red arrow), corrected FPD (FPDc), beating period (BP, green arrow), and amplitude (yellow arrow) were measured. The FPDc was calculated from the FPD and BP measurements. Continuous and representative single electrode waveforms demonstrate the methods used for analysis. (B–E) Heatmaps showing FPDc changes in response to 12 ion channel blockers across normal (iCell, CMC-011, PB01, TnR-4F) and inherited arrhythmia iPSC-CM lines (LQTS: DPHCi01, DPHCi08, DPHKi20; BrS: DPHCi15, DPHKi09; n ≥ 3). (B) K^+ channel blockers. High- torsades de pointes (TdP) risk drugs induced significant FPDc prolongation across cell lines. (C) Na^+ channel blockers. High-TdP risk drugs prolonged FPDc, while low-TdP-risk mexiletine showed minimal effects except in DPHCi08 (red arrow). (D) Ca^{2+} channel blockers. BrS cell lines exhibited higher sensitivity, with marked FPDc changes observed at low concentrations for nifedipine and nitrendipine (blue arrows). (E) Multichannel blockers. Quinidine caused greater FPDc changes in BrS and LQTS lines (black arrow), whereas ranolazine induced greater changes in FPDc at relatively lower concentrations in DPHCi15 cells (purple arrows). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Among dopamine receptor ligands, high-TdP-risk drugs, such as domperidone and droperidol, induced concentration-dependent FPDc changes without clear cell line-specific trends (Fig. 5A). Chlorpromazine and clozapine induced complete beating cessation in LQTS cell lines (red arrow) at high concentrations, indicating that some LQTS-derived cardiomyocytes may be at increased risk for drug-induced repolarization impairment.

Histamine receptor ligands showed variable effects. Astemizole caused pronounced FPDc changes, while terfenadine induced complete beating cessation at high concentrations, resulting in the loss

Table 1
Drug profiles with targets, concentrations, and TdP risk classification.

Type	Target	Drug name	C_{max} (μM)	Drug Concentration (μM)				TdP risk Classification [30]	Credible Meds [31]	Data type
Ion channel blocker	K^+ channel	Azimilide	0.07	0.01	0.1	1	9	High	No report	Testing
	K^+ channel	Sotalol	15	0.1	1	10	100	High	Known risk of TdP	Training
	Na^+ channel	Disopyramide	0.7	0.1	1	10	100	High	Known risk of TdP	Testing
	Na^+ channel	Mexiletine	2.5	0.1	1	10	100	Low	Not classified	Training
	Ca^{2+} channel	Bepridil	0.03	0.01	0.1	1	10	High	Known risk of TdP	Training
	Ca^{2+} channel	Verapamil	0.05	0.001	0.01	0.1	1	Low	Not classified	Training
	Ca^{2+} channel	Diltiazem	0.13	0.01	0.1	1	10	Low	Conditional risk of TdP	Training
	Ca^{2+} channel	Nifedipine	0.008	0.001	0.01	0.1	1	Low	Not classified	Testing
	Ca^{2+} channel	Nitrendipine	0.003	0.01	0.03	0.1	0.3	Low	No report	Testing
	Ca^{2+} / K^+ channel	Ibutilide	0.1	0.0001	0.001	0.01	0.1	High	Known risk of TdP	Testing
	Na^+ / Ca^{2+}/ K^+ channel	Ranolazine	1.9	0.1	1	10	100	Low	Conditional risk of TdP	Training
	Na^+ / K^+ channel	Quinidine	3	0.95	3	9.5	30	High	Known risk of TdP	Training
Dopamine receptor antagonist	D2 / D3 Receptor	Domperidone	0.02	0.003	0.03	0.3	3	Intermediate	Known risk of TdP	Testing
	D2 Receptor	Droperidol	0.02	0.03	0.1	0.3	1	Intermediate	Known risk of TdP	Testing
	D2 / D3 Receptor	Pimozide	0.0004	0.001	0.003	0.01	0.03	Intermediate	Known risk of TdP	Testing
	D1 / D2 Receptor	Chlorpromazine	0.0345	0.095	0.3	0.95	3	Intermediate	Known risk of TdP	Training
	D2 Receptor	Clozapine	0.07	0.1	0.3	0.9	3	Intermediate	Possible risk of TdP	Testing
Histamine receptor antagonist	H1 Receptor	Astemizole	0.0003	0.0001	0.001	0.01	0.1	Intermediate	Known risk of TdP	Testing
	H1 Receptor	Terfenadine	0.0003	0.001	0.01	0.1	1	Intermediate	Known risk of TdP	Training
	H1 Receptor	Loratadine	0.00045	0.001	0.003	0.01	0.03	Low	Not classified	Testing
Serotonin receptor ligand	5-HT3 Receptor antagonist	Ondansetron	0.37	0.03	0.3	3	30	Intermediate	Known risk of TdP	Training
	5-HT4 Receptor agonist	Cisapride	0.0026	0.003	0.01	0.03	0.1	Intermediate	Known risk of TdP	Training
	5-HT2A Receptor antagonist	Risperidone	0.0018	0.003	0.01	0.03	0.1	Intermediate	Conditional risk of TdP	Testing
Others	Tyrosine kinase inhibitor	Vandetanib	0.3	0.01	0.1	1	10	High	Known risk of TdP	Testing
	Macrolide antibiotics	Clarithromycin	1.2	0.1	1	10	100	Intermediate	Known risk of TdP	Testing
	Estrogen receptor ligand	Tamoxifen	0.022	0.1	0.3	0.9	3	Low	Possible risk of TdP	Testing
Ion channel blocker	K^+ channel	Amiodarone	0.0008	0.1	1	10	40	Not classified	Not classified	
Others	Antiviral agent	Remdesivir	9	3	10	30	100	Not classified	Not classified	–

of detectable field potential signals (Fig. 5B). The cardiotoxicity risks associated with astemizole and terfenadine align with their withdrawal from the market due to high TdP risk. Loratadine, a low-TdP-risk drug, exhibited no significant effects across cell lines [35].

For serotonin receptor ligands, ondansetron caused dose-dependent FPDc changes, with greater alterations in arrhythmic cell lines than in normal cells (dotted red line in Fig. 5C). These findings are consistent with reports linking high doses of ondansetron to QT prolongation, though its use at lower doses in non-cardiac patients poses minimal cardiotoxic risk [36]. Cisapride exhibited minimal FPDc changes, limited to specific cell lines (blue arrow in Fig. 5C), while risperidone showed no significant change in all cell lines.

Among other non-cardiovascular drugs (Fig. 5D), vandetanib, a tyrosine kinase inhibitor, demonstrated dose-dependent FPDc changes across all cell lines, reflecting its known association with QT interval prolongation in clinical settings. Clarithromycin induced minor FPDc changes, with flat repolarization potentials recorded in select LQTS and normal cell lines (black arrow in Fig. 5D). Tamoxifen, an estrogen receptor ligand, caused a distinct absence of FPDc recordings low concentration (0.3 μM) specifically in the DPHCi08 cell line (purple arrow in Fig. 5D).

These findings highlight the varying cardiotoxic responses of normal and inherited arrhythmic iPSC-CMs to non-cardiovascular drugs. Arrhythmic cardiomyocytes exhibited heightened sensitivity, emphasizing the importance of disease-specific evaluations.

Development of an MEA-ML platform for TdP risk classification

In order to predict the risk of drug-induced cardiotoxicity based on electrophysiological responses, a ML–based platform was developed that integrates MEA data from both normal and inherited arrhythmic iPSC-CMs (Fig. 6). The platform was designed to clas-

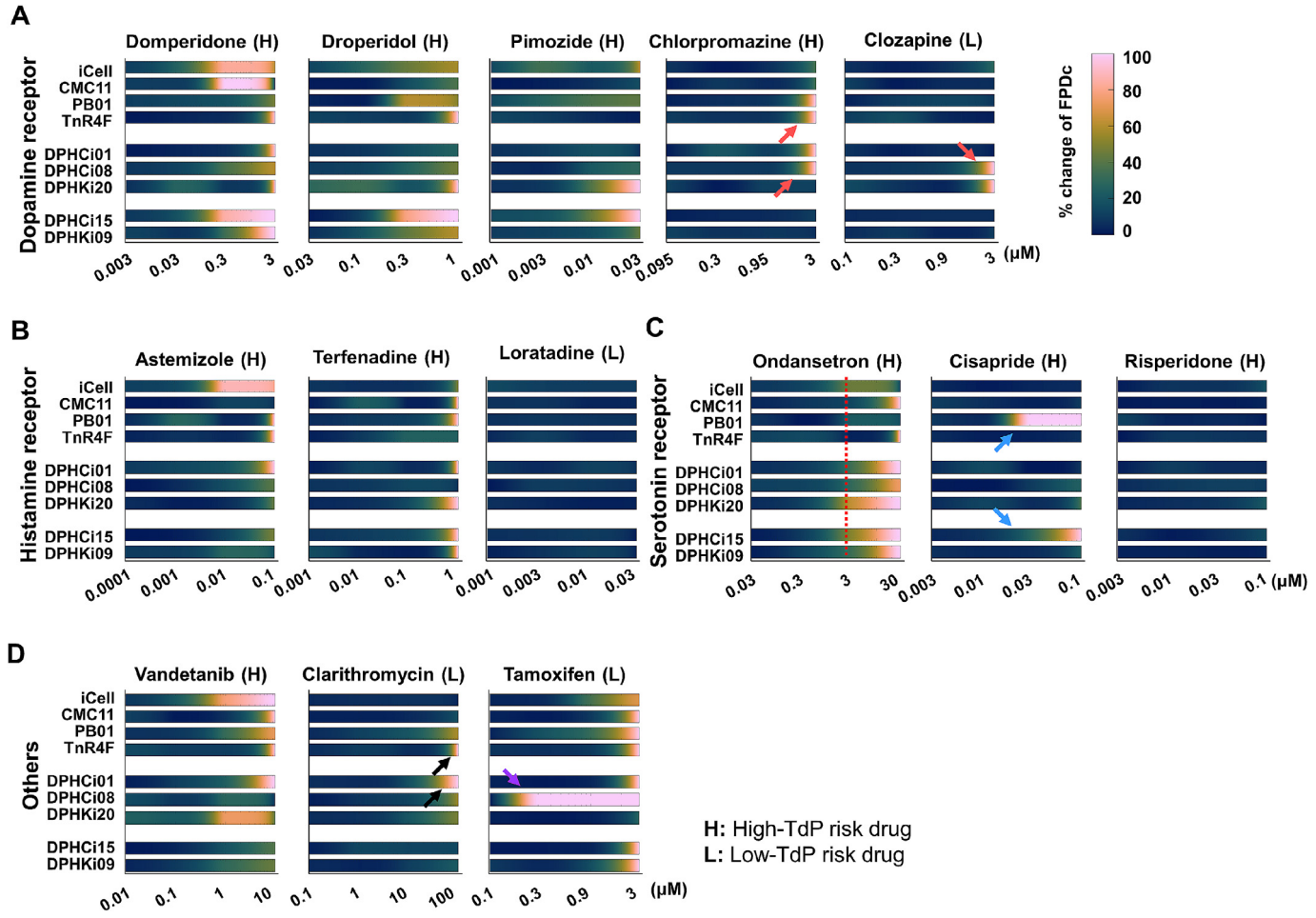


Fig. 5. Drug-induced corrected field potential duration (FPDc) changes in cardiomyocytes for non-cardiovascular drugs. Heatmap images of FPDc changes rate in response to non-cardiovascular drugs across normal and inherited arrhythmia iPSC-CM lines (n ≥ 3). (A) Changes in FPDc induced by dopamine receptor ligands. High-torsades de pointes (TdP)-risk drugs caused concentration-dependent increases in FPDc across all cell lines. Chlorpromazine showed repolarization instability at high concentrations in specific normal and LQTS cell lines (red arrows). Clozapine caused similar effects in DPHCi08 cells (red arrow). (B) FPDc responses to histamine receptor ligands. Astemizole induced consistent FPDc increases across all cell lines, while terfenadine caused flat repolarization potentials at high concentrations. Loratadine did not significantly affect FPDc in any cell line. (C) Effects of serotonin receptor ligands. Ondansetron caused marked, dose-dependent increases in FPDc, with a significant effect observed in LQTS and BrS cell lines at high concentrations (dotted red line). Cisapride induced FPDc alterations in PB01 and DPHCi15 cell lines only (blue arrows). (D) Responses to other non-cardiovascular drugs. Clarithromycin induced FPDc changes at high concentrations, with flat repolarization potentials observed in TnR4F and DPHCi01 cell lines (black arrows). Tamoxifen did not induce FPDc changes, except in the DPHCi08 cell line at 0.3 μM (purple arrow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sify drugs into high- and low-TdP risk categories using systematic preprocessing, feature extraction, and multiple classifier models.

The analytical workflow consisted of five key steps (Fig. 6A). Firstly, MEA drug response data was screened to select clinically meaningful concentration ranges (10^{-2} to 10 × Cmax), encompassing sub-therapeutic to supratherapeutic levels. This ensured that observed effects reflected plausible clinical exposure levels, while avoiding concentrations too low to elicit measurable responses or too high to cause non-specific toxicity (Fig. S3A). Then, to mitigate variability across replicates and cell lines, we applied data augmentation by averaging MEA parameters (FPD, FPDc, beat period, and amplitude) across the selected four concentrations per drug. This process enhanced dataset uniformity and increased model robustness (Fig. S3B). The dataset, comprising 26 drugs, was then divided into a training group (11 drugs) and a testing group (15 drugs) for the purposes of supervised learning. Three machine learning algorithms—artificial neural network (ANN), random forest (RF), and XGBoost—were implemented to evaluate TdP risk based on drug-induced changes in MEA parameters. Finally, the performance of the model was assessed across multiple cell line scenarios to identify optimal configurations for downstream applications.

The ANN model was implemented in Keras with two hidden layers and a sigmoid classifier, using the four MEA parameters as input features (Fig. 6B). The RF model, on the other hand, employed an ensemble of decision trees to enhance prediction robustness. Each tree independently classified the input, and the final classification was based on majority voting, reducing the influence of outliers and enhancing generalizability (Fig. 6C). The XGBoost model, founded on gradient boosting, trained each decision tree sequentially, with each subsequent tree correcting the errors of its predecessor. Node splitting was guided by an objective loss function, and regularization was applied to prevent overfitting. This approach enabled the precise optimization of local decision boundaries while preserving global performance (Fig. 6D).

Consequently, the amalgamation of these models resulted in a multilayered prediction framework, which was capable of identifying TdP risk signatures embedded in MEA responses across

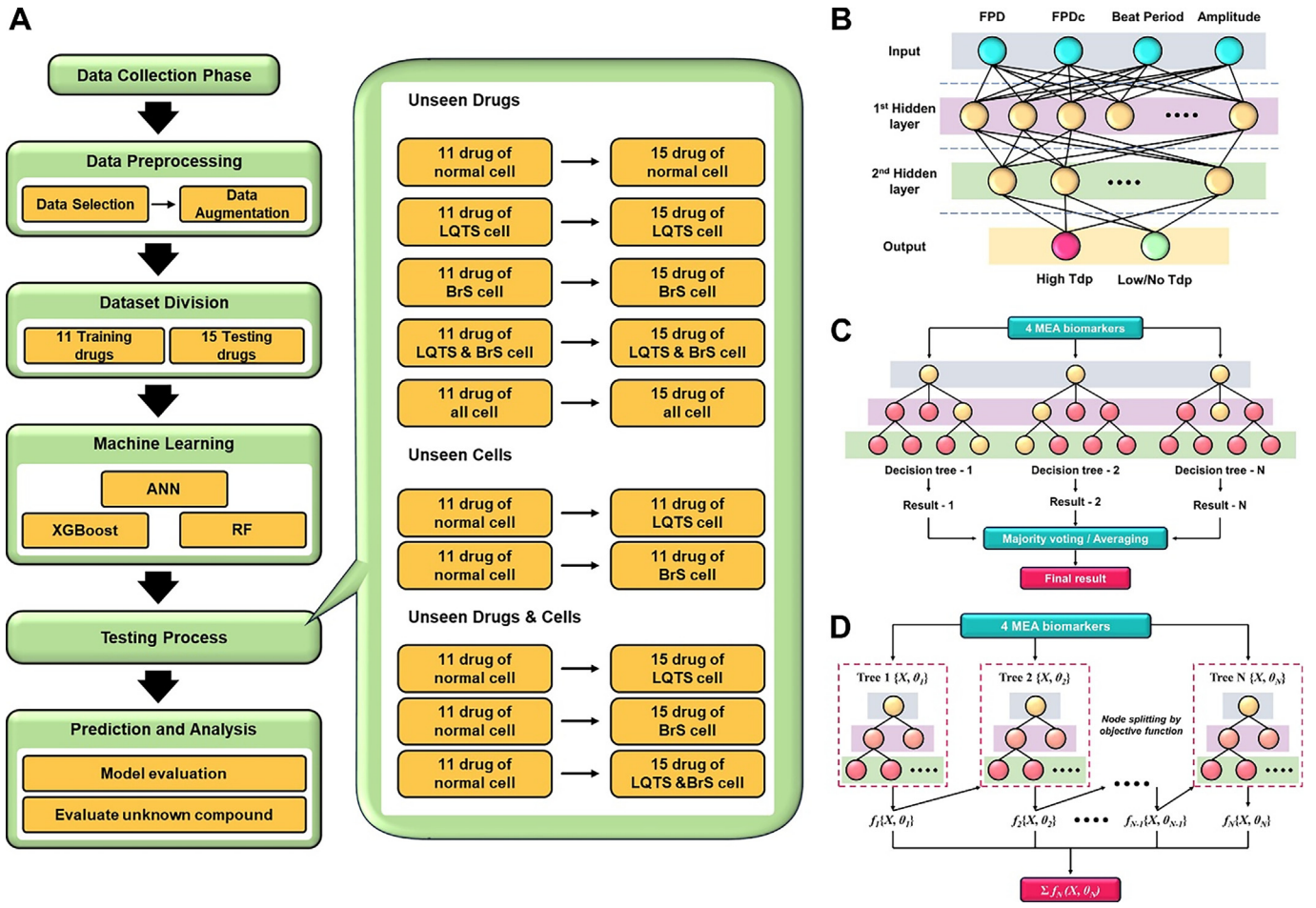


Fig. 6. Machine learning (ML) workflow for drug-induced TdP risk prediction using MEA data. (A) Workflow for the development and evaluation of ML models for TdP risk prediction. The process includes data collection, preprocessing (selection and augmentation), and dataset division into training (11 drugs) and testing groups (15 drugs). ML algorithms, artificial neural network (ANN), random forest (RF), and XGBoost, were trained and tested with various unseen scenarios, including drugs, cell types (normal, LQTS, and BrS), and combinations of drugs and cell types. Unseen test conditions enabled evaluation of the models' generalizability for predicting the TdP risk of unknown compounds. (B) Architecture of the ANN model for TdP risk classification. Four MEA biomarkers (FPD, FPDc, beat period, and amplitude) were used as input features, with two hidden layers for feature mapping and one output layer for binary classification into high or low/no TdP risk categories. (C) Structure of the RF model. Multiple decision trees were constructed based on the four MEA biomarkers, and predictions from individual trees were combined through majority voting or averaging to provide a final TdP risk classification. This ensemble approach improved model reliability and predictive accuracy. (D) Structure of the XGBoost model. Gradient boosting was employed to train decision trees sequentially, with each tree optimized to minimize prediction errors from prior iterations. Node splitting using an objective function enabled the model to refine predictions and improve accuracy in TdP risk estimation.

different arrhythmia disease. The trained platform was subsequently employed to evaluate unclassified compounds and to validate the prediction accuracy in disease-specific cell lines.

Evaluation of ML model performance and classification of drug-induced TdP risk

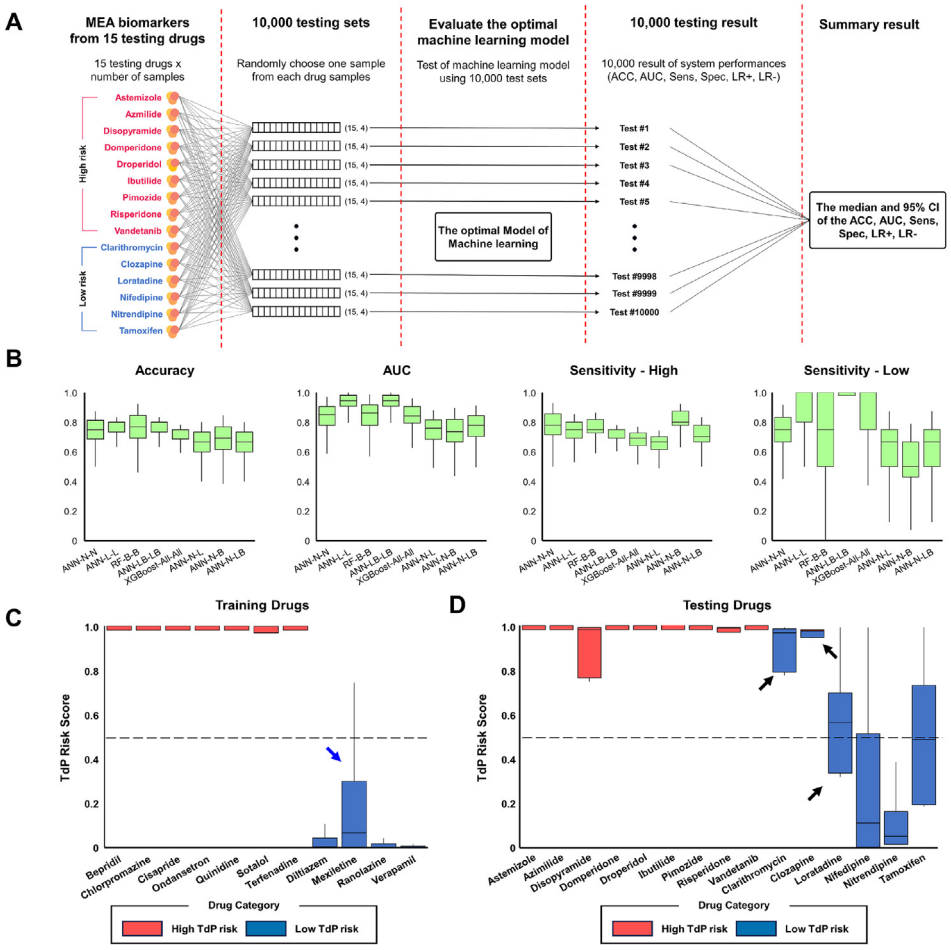
We evaluated the performance of three machine learning models—ANN, RF, and XGBoost—by categorizing the TdP risk of drugs using MEA data from normal and inherited arrhythmic iPSC-derived cardiomyocytes (Fig. 7). Eight classification scenarios were tested: five for training and testing within the same cell line group (normal, LQTS, BrS, inherited arrhythmia, or all lines) and three for training on normal cell lines and testing on arrhythmia cell lines. For each scenario, the models were trained using 11 compounds and tested on 15 additional compounds (Table 1).

To assess performance, 10,000 tests were conducted using the 15-drug test set, each involving all four MEA parameters (Fig. 7A). Performance metrics—accuracy, sensitivity, specificity, area under the curve (AUC), F1 score, positive likelihood ratio (LR^+), and negative likelihood ratio (LR^-), were calculated as described in Equations 1–5 in the detailed methods section of the supplemental material.

The ANN model consistently demonstrated the highest performance across scenarios. The RF model performed best for BrS-specific cases, while XGBoost was most effective when trained on the complete dataset. Among single-line training scenarios, the highest AUC (0.94) was observed when the ANN model was trained and tested on LQTS or BrS lines. Accuracy across these scenarios ranged from 0.75 to 0.80, with AUC values of 0.84–0.94 (Fig. 7B, Table S4).

In contrast, models trained solely on normal cell lines showed reduced accuracy (0.67–0.69) when applied to arrhythmic lines, with corresponding AUC values between 0.74 and 0.78. This highlights the limitations of using healthy-cell-trained models to evaluate drugs in disease-specific backgrounds and emphasizes the importance of incorporating disease-matched training data.

To validate classification accuracy, we plotted the distribution of predicted TdP scores for both training and testing drugs using one of the best-performing models (ANN trained on LQTS cells, red arrow in Fig. 7B). The ANN model correctly classified all high-TdP-risk training drugs with median scores approaching 1,



E

Category	Drug	Normal (4 Cell lines)	LQT (3 Cell lines)	BrS (2 Cell lines)
High TdP Risk	Astemizole	2.0	2.0	4.0
	Azimilide	1.8	3.0	3.5
	Bepridil	2.8	3.7	2.5
	Chlorpromazine	2.0	3.7	3.0
	Cisapride	2.8	2.3	3.5
	Disopyramide	3.0	3.0	4.0
	Domperidone	3.0	2.3	4.0
	Droperidol	2.3	3.0	2.5
	Ibutilide	2.0	3.3	3.0
	Ondansetron	2.8	2.7	2.5
	Pimozide	1.8	2.3	4.0
	Quinidine	3.0	4.0	4.0
	Risperidone	1.3	2.7	1.0
	Sotalol	3.0	3.3	3.5
	Terfenadine	2.0	1.7	3.5
	Vandetanib	2.8	3.7	1.5
Low TdP Risk	Clarithromycin	1.8	1.7	2.0
	Clozapine	1.5	2.7	3.0
	Diltiazem	2.8	3.0	3.0
	Loratadine	1.0	1.7	3.0
	Mexiletine	2.0	2.7	3.0
	Nifedipine	2.3	2.7	3.0
	Nitredipine	3.3	2.7	4.0
	Ranolazine	2.0	3.0	3.0
	Tamoxifen	2.0	2.3	3.0
	Verapamil	3.3	3.0	3.5

TdP risk level: 4, 3, 2, 1, 0

while most low-risk drugs had scores near zero—except mexiletine, which showed slightly higher scores but remained below the 0.5 threshold (blue arrow in Fig. 7C). Among the 15 test compounds, most predictions aligned with known risk profiles. Disopyramide exhibited a slightly lower median score (0.991), and clarithromycin, clozapine, and loratadine—classified as low-risk in prior studies—were predicted as high-risk with median scores > 0.5 (black arrow in Fig. 7C). These findings suggest that some drugs considered low-risk may elicit arrhythmogenic responses, particularly in cells with inherited arrhythmia backgrounds. Supporting clinical evidence includes the association of clarithromycin and clozapine with QT prolongation and increased cardiovascular risk [37–39].

To further evaluate model generalizability, the ANN model trained on normal cells was used to predict responses in LQTS and BrS lines. The predicted TdP risk often diverged from clinical expectations, especially for low-risk drugs, which were frequently reclassified as high-risk in disease-specific lines (Fig. S4A–B). This reinforces the need for stratified modeling.

Finally, to quantify drug-induced effects by concentration, we calculated FPD20, FPDc20, BP20, and Amp20—the lowest concentrations yielding ≥ 20 % changes in MEA features—across four dosing levels. These values were converted into discrete risk levels (0–4) based on the concentration at which the threshold change occurred (Fig. 7E). Risk scores were consistently higher in LQTS and BrS lines compared to normal lines.

These results underscore the utility of disease-stratified ML models in improving cardiotoxicity risk prediction and demonstrate the importance of incorporating disease-relevant cell types in predictive frameworks. The observed responses closely reflect known clinical variability in arrhythmia susceptibility.

Application of the predictive platform to clinically relevant unclassified compounds

The accuracy of the predictive system was validated for two new drugs, amiodarone and remdesivir, with unclassified risk levels (Fig. 8). Amiodarone is an antiarrhythmic agent and remdesivir is one of the Emergency Use Authorization drugs for Coronavirus disease 2019 (COVID-19) pandemic. Although these drugs do not have high- or low-risk classifications, all have documented cases of cardiotoxicity. Amiodarone has been documented to induce QTc prolongation and TdP [40,41], while remdesivir has been linked to changes in heart rate and cardiac arrest during the COVID-19 pandemic [42]. To evaluate the potential cardiotoxicity of amiodarone and remdesivir, which have been associated with adverse cardiac events in both healthy individuals and patients with inherited arrhythmias, we employed MEA data to quantify the rates of change in FPDc with treatment (Fig. 8A).

Both amiodarone and remdesivir elicited concentration-dependent changes in the FPDc. The MEA results were used to evaluate the cardiotoxicity risk associated with each drug (Fig. 8B). The TdP risk scores in the normal, LQTS, and BrS cells were all high, with a minimum of 0.5 (data not shown). The LQTS cell line was associated with the greatest model performance, exhibiting a high risk with a minimum value of 0.8 (Fig. 8B). In Fig. 7D, we evaluated the cardiotoxic sensitivity for the drugs according to cell line using the FPD20, FPDc20, BP20, and Amp20 parameters. Based on this analysis we also found that both drugs exhibited a heightened risk in LQTS and BrS cell lines relative to normal cell lines which was further substantiated the elevated cardiotoxic sensitivity of the inherited arrhythmia cell lines (Fig. 8C). These findings demonstrate the platform's capability to detect hidden cardiotoxic risk in genetically susceptible populations, even for clinically unclassified compounds.

Discussion

This study used hiPSC-CMs and MEA technology to compile a database of cardiomyocyte responses to 28 drugs, including 26 with a classified cardiotoxicity risk [43–45] and two with reports of cardiotoxicity risk, but no official classification, predicted using an artificial intelligence-powered ML model. By integrating multichannel MEA data with ML, we achieved a clinically aligned and disease-specific framework for cardiotoxicity prediction. In particular, the platform also reclassified several compounds-including non-cardiac drugs previously deemed low risk-highlighting its potential for uncovering hidden proarrhythmic liabilities.

The hiPSC-CMs offer an ethically sustainable and genetically representative model for studying patient-specific electrophysiological responses, addressing limitations in accessibility, cost, and variability associated with primary human cardiac tissues. Their close genomic correspondence to donor cells makes them particularly attractive for evaluating individualized drug responses [46]. However, modeling inherited arrhythmia remains challenging: despite harboring pathogenic mutations, not all hiPSC-CM lines recapitulate expected disease phenotypes. In this study, iPSC-CMs derived from patients with LQTS and BrS underwent careful selection based on both their genetic mutations and electrophysiological characteristics. Notably, three out of eight mutation-carrying lines failed to exhibit LQTS- or BrS-like features, underscoring the necessity of functional validation beyond genotyping alone. This highlights the variable penetrance and expressivity observed in inherited cardiac diseases and emphasizes the importance of combining genotypic and phenotypic screening when building disease-relevant *in vitro* models [47].

The MEA system is a non-invasive way of evaluating the electrophysiological properties of different ion channels acting on the FP of cardiomyocytes. It is an enhanced methodology for cardiotoxicity evaluation compared with traditional single-channel or cell line-based models. Previous studies predominantly employed hERG assays to predict the arrhythmia risk of nascent pharmaceuticals, although this single-channel approach may overlook the potential of many drugs for inducing unanticipated arrhythmias

Fig. 7. Machine learning (ML)-based TdP risk assessment. (A) Scheme of performance metric analysis for 15 testing drugs using machine learning models. Three different ML models were employed over 10,000 iterations to obtain the median values and 95 % confidence intervals (CI) for accuracy (ACC), area under the curve (AUC), sensitivity, specificity, positive likelihood ratio (LR^+), and negative likelihood ratio (LR^-). (B) Boxplots of performance metrics (accuracy, AUC, sensitivity, and specificity) for ANN, RF, and XGBoost models trained and tested across eight scenarios. ANN demonstrated the highest AUC (0.94) for LQTS and inherited arrhythmia cell lines, while RF and XGBoost performed best for BrS and combined cell lines, respectively. (C) TdP risk scores for training drugs in ANN model trained on the LQTS database. High-TdP-risk drugs consistently scored ≥ 0.5 (red bars), while mexiletine showed variability among low-TdP-risk drugs (blue arrow). (D) TdP risk scores for testing drugs using an ANN model trained on the LQTS database. High-TdP-risk drugs maintained scores ≥ 0.5, while clarithromycin, clozapine, and loratadine showed elevated scores in diseased cell lines (black arrows), indicating heightened sensitivity in LQTS and BrS cells. (E) Heatmap of drug risk levels based on FPD20, FPDc20, BP20, and Amp20 across cell lines. LQTS and BrS cell lines exhibited higher risk levels, with discrepancies noted for mexiletine and ranolazine, classified as high-risk in diseased cells despite their low-risk profiles. This highlights the influence of cardiac conditions on TdP risk assessment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

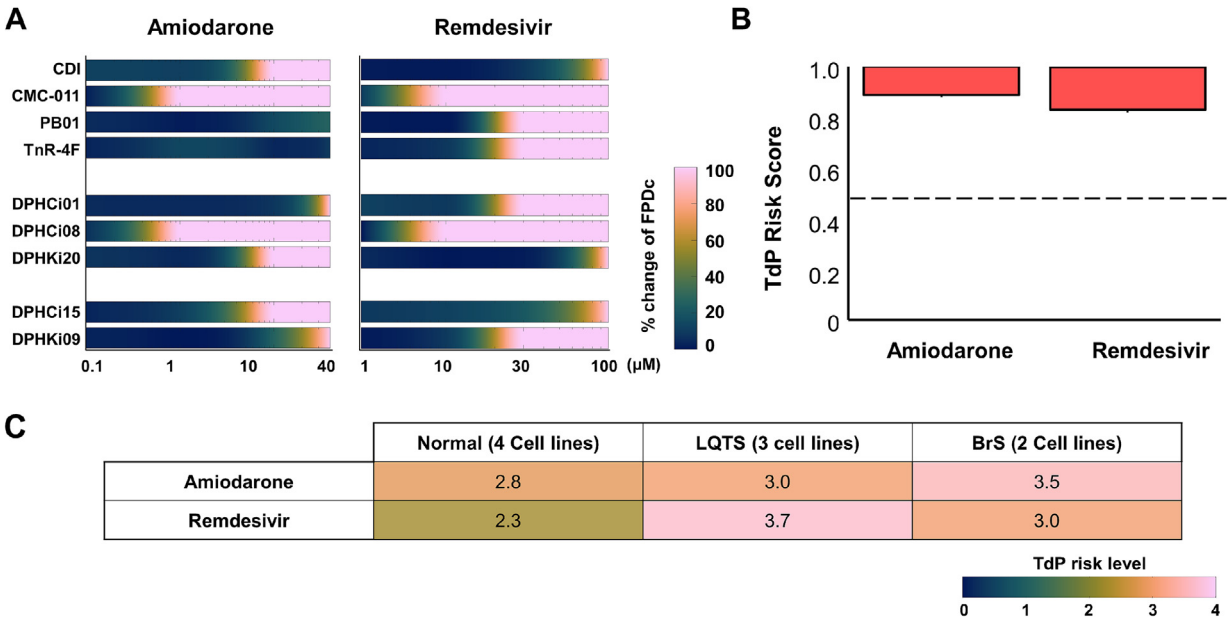


Fig. 8. Risk analysis of uncategorized drugs, amiodarone and remdesivir, using MEA data and ML-based models. (A) Heatmap illustrating the percentage change in FPDc for amiodarone and remdesivir across nine cell lines, including normal, LQTS and BrS. Both drugs demonstrated concentration-dependent FPDc prolongation, with greater changes observed in LQTS and BrS cell lines. (B) Boxplot of TdP risk scores for amiodarone and remdesivir across all cell lines. Both drugs exhibited uniformly high risk, with scores exceeding the 0.5 threshold across normal and arrhythmia cell lines. The LQTS cell line demonstrated the highest sensitivity, with scores above 0.8. (C) Heatmap of TdP risk levels (0–4) based on FPD20, FPDc20, BP20, and Amp20 parameters across normal, LQTS, and BrS cell lines. Both drugs were associated with higher risk levels in LQTS and BrS cell lines compared with that of normal cell lines, emphasizing the increased sensitivity of arrhythmic models to cardiotoxicity.

[5]. In contrast, the present study employed the MEA system to comprehensively evaluate the impact of diverse ion channels and more precisely reflect the actual arrhythmia risk. Furthermore, as hiPSC-CMs are increasingly utilized to evaluate the cardiotoxicity of pharmaceutical agents, various methodologies have been developed to integrate them into ML models, representing the latest advancements in bioinformatics. We effectively predicted the risk of drugs based on biomarkers derived from the drug responses of hiPSC-CMs based on the changes in Ca^{2+} signals [48] and action potential [10] using a RF model and conventional neural networks, achieving AUC values of up to 0.95. We used two distinct methodologies: MEA, and ML models. The latter employed a consistent algorithm free from human bias to analyze a substantial quantity of data, thereby enabling consistent and TdP risk prediction. The implementation of a mechanistic procedure-based system for evaluating the cardiotoxicity of pharmacological agents can reduce the influence of experimenter or rater error and minimize human intervention. This approach may enhance the reliability of drug risk prediction as well as standardization of drug development and rollout. However, MEA signals reflect only extracellular field potentials and may not fully capture complex arrhythmogenic mechanisms, such as spatial reentry or local conduction block. These limitations have been previously discussed by Elkins et al. [49], who emphasized the importance of combining complementary methodologies. In particular, the integration of mechanistic in silico models and data-driven machine learning approaches was suggested to improve the comprehensiveness of cardiac safety assessment.

This study made several important findings. K^+ (e.g., azimilide and sotalol) and Ca^{2+} channel blockers (verapamil and nifedipine) are clinically arrhythmogenic drugs. We found that these drugs caused FPDc prolongation in inherited arrhythmia cell lines (LQTS and BrS). These findings indicated that K^+ channel blockers may be associated with an increased risk of arrhythmia in LQTS cell lines. Conversely, the BrS cell lines were particularly sensitive to Ca^{2+} channel blockers, suggesting that the use of Ca^{2+} blockers should be limited in patients with BrS. The findings align with real-world electrophysiological responses, which suggests that cardiotoxicity evaluations using disease-specific cell line models can substantially enhance patient safety. Mexiletine is an Na^+ channel blocker that shortens the QTc interval in LQTS cell lines by inhibiting excessive Na^+ currents caused by *SCN5A* mutations [50,51]. The DPHCi08 cell line carrying a *KCNQ1* mutation with Long QT syndrome type 1 (*LQT1*) characteristics showed high sensitivity to mexiletine. Although mexiletine is low risk, it has been associated with FPD prolongation and signal attenuation resulting from hERG channel inhibition at elevated concentrations. This demonstrates the differences in drug responses based on specific gene variants, such as *LQT1*, and emphasizes the need for considering genetic background in drug selection. Disopyramide is a class 1A antiarrhythmic drug that inhibits Na^+ channels, thereby suppressing ventricular tachycardia [52]. In the present study, disopyramide had a low risk, which is attributed to an arrhythmia-stabilizing effect associated with Na^+ channel blockade. This reflects the usefulness of ML-based methodologies for validating the effects of low-risk drugs on cardiotoxicity in conventional clinical evaluations based on the electrical properties of authentic patient cardiomyocytes. Importantly, these patient-derived iPSC-CMs serve as functional biomaterials that recapitulate disease-relevant electrophysiology, enabling interpretable, high-throughput evaluation of compound-specific risk.

Certain non-cardiovascular medications have been implicated in unexpected cardiotoxic responses. Clarithromycin and clozapine, previously classified as low risk for QT prolongation and TdP, were reclassified as high-risk based on findings from this study. Both drugs demonstrated high-risk profiles in LQTS cell lines, characterized by FPDc prolongation and FP signal attenuation. Additionally, clozapine has been reported to affect QT intervals through hERG channel inhibition in prior studies. These results collectively suggest a need to reevaluate existing criteria for cardiotoxicity risk assessment [38,53], and underscore the need for carefully evaluating the cardiotoxicity of both cardiovascular and

non-cardiovascular pharmaceuticals and the importance of monitoring patient medications.

The AI-powered ML model demonstrated superior accuracy in predicting cardiotoxicity compared to traditional human-based analytics. A comparative analysis of multiple ML algorithms, ANN, RFs, and XGBoost, highlighted disease-specific optimization patterns. While ANN consistently achieved the highest accuracy overall, especially when trained on inherited arrhythmia cell lines, while RF demonstrated superior performance in BrS-specific scenarios. XGBoost showed improved generalizability across mixed-disease datasets, suggesting its applicability to scenarios involving heterogeneous patient populations or limited training data. The changes in FPDc based on the MEA was a particularly useful metric for predicting arrhythmia risk, establishing a new paradigm for cardiotoxicity assessment. The proposed modeling framework can be used to predict cell-specific cardiotoxicity for patients with different genetic backgrounds and substantially advance patient-specific drug evaluations. However, AI analytics still heavily rely on the quantity and quality of data. Therefore, further correlations with a diversity of real-world clinical datasets are needed to validate the generalizability of our approach.

Conclusion

This study highlights the potential of integrating patient-specific hiPSC-derived cardiomyocytes with MEA-based electrophysiological profiling and ML for cardiotoxicity assessment. By capturing disease-dependent drug responses, we established a clinically relevant risk prediction framework that can enhance preclinical safety evaluation and support personalized cardiovascular care.

Compliance with ethics requirements

This study was conducted in accordance with the ethical guidelines of the Declaration of Helsinki and approved by the Institutional Review Board/Ethics Committee of Seoul St. Mary's Hospital (No. KC16TNMI1023).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jare.2025.07.030.

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